

The Closure Loop Gas Architecture: Endogenous Parameter Derivation, Information-Geometric Duals, and Cosmological Implications

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Introduction: The Ontological Inversion and Primitive Substrate

The trajectory of contemporary theoretical physics has reached a critical juncture characterized by the irreconcilable structural fracture between the deterministic, smooth manifold geometry of general relativity and the probabilistic, discrete operational mechanics of quantum field theory. Traditional unification programs, ranging from multi-dimensional string theory to loop quantum gravity, have predominantly adhered to a substance-based ontology. These frameworks attempt to quantize pre-existing gravitational fields or map string excitations onto a pre-existing spacetime background, yet they consistently encounter insurmountable mathematical difficulties regarding the Measurement Problem and the spontaneous emergence of spacetime itself.

The Closure Loop Gas (CLG) program enacts a rigorous ontological inversion.¹ It postulates that physical reality is not a pre-existing geometric container of finished objects governed by external laws, but rather a recursively self-compiling computational closure process. In this paradigm, the universe operates as an

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active, dynamic manifold of partial prefixes continually tested for admissible continuation. The conventional ontological sequence—vacuum $\rightarrow$ matter $\rightarrow$ law $\rightarrow$ computation—is entirely rejected. In its place, the framework proposes a sequence dictated by informational logic: difference $\rightarrow$ active field $\rightarrow$ wave logic $\rightarrow$ matter $\rightarrow$ computation as a local harness.¹

Three co-present primitives form the necessary, bedrock conditions for the emergence of any physical structure within the Open Distinction Field ¹:

Primitive	Symbol	Foundational Role
Difference / Gap	Δ	The absolute condition of distinguishability. Without differentiation, nothing exists to be measured. This is prior to number, object, and coordinate.
Touch / Interface	Γ	The possibility of relation and the initiating boundary trigger for operational closure loops.
Conservation / Invariant	I	The condition of persistence. Without structural invariance, no patterned information survives entropic decay.

Under these primitives, a completed closure loop constitutes the minimal local audit log of a resolved physical event, governed by the operational sequence: $\Gamma \rightarrow K \rightarrow \Psi \rightarrow T \rightarrow R \rightarrow \Gamma'$.¹ In this sequence, Γ is the initiating contact boundary, K represents local kinematic resolution, Ψ is the stored

closure record, T denotes the trace readout channel, R is the resolved state, and Γ' establishes the subsequent boundary written by the completed event.¹ Matter, therefore, is not a primary substance but a stabilized closure trace—a locally held result of recursive continuation.

The CLG program bifurcates its analysis into two inseparable channels. The Shape Channel dictates the invariant geometric grammar, defining what classes of objects can mathematically exist. Concurrently, the Value Channel performs the continuous audit to test whether a current structural rendering is mathematically admissible.¹ Previous iterations of the CLG framework utilized Planck-natural parameters inserted by hand to validate the Value Channel. However, the current solve-state of the program achieves total endogenous parameter derivation.¹ By forcing the gravitational structure, defining the minimal dual action, and calculating a parameter-free cosmic abundance discriminator (the Y-discriminant), the framework systematically falsifies thermal equilibrium models of the early universe and bridges the macro-geometric scale with discrete information theory.

Discrete Substrate Dynamics: Field Mathematics and Incremental Growth

To understand how a continuous macroscopic geometry emerges, it is necessary to examine the discrete computational mechanics operating at the substrate level. The local readouts of state—often generalized in localized nodes as "hot" or "cold" regimes—are merely thresholded projections of a deeper, mathematically rigorous admissibility field.¹

The Variational Energy Functional and Graph Laplacian

At the layer immediately above simple binary readouts, the structural evolution of the universe is governed by field mathematics rather than label mathematics.¹ The universal condition that makes comparison possible relies on a globally available distinguishability field, $\mathcal{D}$, and a Comparator field, $\mathcal{C}$.¹ Within this architecture, an admissibility potential field is defined as $\Phi(x, t)$, representing local compile pressure. The visible macroscopic states are defined entirely by thresholding this field: a state is "hot" if $\Phi > \theta_h$ and "cold" if $\Phi < \theta_c$.¹ The true dynamics are driven by gradient, curvature, source/sink mechanics, and threshold hysteresis.¹

Defining a state field $u(x, t)$, the system minimizes an energy or burden functional $F[u]$ ¹:

$$F[u] = \int \left(\frac{\alpha}{2} \|\nabla u\|^2 + V(u) + \lambda C(u, N_x) \right) dx$$

The components of this functional translate computational logic into geometric tension. The term $||\nabla u||^2$ imposes a shape tension or boundary cost, $V(u)$ dictates the local basin preference, and $C(u, N_x)$ represents neighbor coupling, which is equivalent to runtime compatibility. The actual motion of the substrate field is driven by the variation ¹:

$$\partial_t u = -\kappa \frac{\delta F}{\delta u} + J(x, t)$$

where $J(x, t)$ acts as an external input or informational source. In this variational context, a "hot" region signifies that the local variation is strongly positive and structurally unresolved, demanding computational cycles. A "cold" region implies the local site is relaxed, sink-like, or permanently committed.¹ The energetic cost of committing these relations manifests as physical heat, Θ , which is proportional to the rate of change of the rendering cost density $\mathcal{R}_\Theta \propto d\mathcal{R}/dt$.¹

When discretized into a graph geometry, this continuous variation translates into a graph Laplacian system ¹:

$$\partial_t u_i = -\kappa \left(\sum_j L_{ij} u_j + V'(u_i) + \lambda \partial_i C \right) + J_i$$

where L_{ij} is the graph Laplacian. This mathematical framework explicitly demonstrates that the fundamental driver of physical reality is a Laplacian operator interacting with local potentials and source mathematics. The structural topology Σ_n maps to the value field Φ_{n+1} via the directional operator D_G , which subsequently partitions into the hot/cold boundary defining accepted physical motion: $\Phi_{n+1}(x) = D_G(x)$.¹

Incremental Admissibility and Modular Nibble Growth

If the substrate's resolution of "hot" and "cold" is computationally real, it must operate in discrete increments. The framework models this utilizing a modular arithmetic paradigm, conceptualizing complex structures—such as a 32-bit word—not as pre-existing entities, but as an 8-step base-16 compile grown incrementally from "nibbles".¹ Any structured state X can be defined as a summation of local states:

$$X = \sum_{j=0}^7 16^j x_j, \quad x_j \in \{0, \dots, 15\}$$

When this state X is grown within a broader modular sum $T \equiv S + X \pmod{2^{32}}$, where S represents the summation of already-known environmental contributors, the growth is governed by an exact local law at each nibble j ¹:

$$s_j + x_j + c_j = t_j + 16c_{j+1}$$

Here, t_j is the target nibble, s_j is the known sum, x_j is the unknown, and c_j represents the carry-in from the previous discrete state. This yields the deterministic nibble-grow rule and its required carry propagation:

$$x_j \equiv t_j - s_j - c_j \pmod{16}$$

$$c_{j+1} = \lfloor (s_j + x_j + c_j) / 16 \rfloor$$

The physical interpretation of this arithmetic defines the thermodynamic state of the substrate. In a "cold growth" regime, where there is only one unknown channel within the modular sum, every increment is uniquely determined once the carry-in is known. The admissible set of solutions, A_j , contains exactly one element ($|A_j| = 1$), representing locally unique continuation.¹

Conversely, the "hot frontier" emerges when multiple unknown channels interact, such as $T \equiv S + X + Y \pmod{2^{32}}$. The exact local law expands to incorporate the new variable, resulting in an admissible set of multiple valid continuations¹:

$$A_j = \{(x_j, y_j, c_{j+1}) : s_j + x_j + y_j + c_j = t_j + 16c_{j+1}\}$$

In this multiple-unknown regime, the magnitude of the set is $|A_j| = 16$. The system cannot automatically advance; it requires neighboring structural constraints to cool the frontier until a unique path crystallizes.¹ This substrate behavior naturally defines local heat (H_j) and lock strength (L_j):

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$$H_j = \log_2 |A_j|, \quad L_j = \frac{1}{|A_j|}$$

A heat value of $H_j = 0$ signifies a fully cold, unique path, whereas larger values indicate high branching potential. Words and structures do not appear whole; they crystallize upward from nibble-local acceptance and carry motion, proving that the foundation of reality relies on modular arithmetic actualization.¹

Macro-Geometric Closure and Action Functionals

While graph Laplacians and nibble logic govern the discrete scale, the continuous macroscopic limit must conform to strict geometric constraints to remain physically legible. The macroscopic gravitational law in the CLG framework is not an ad hoc postulate; it is the unique mathematical outcome of imposing four structural persistence constraints on the geometric side of the field equations.¹

To sustain a gap-first ontology, the governing law must satisfy locality and differentiability (preventing non-local rendering updates), general covariance (eliminating preferred coordinate frames), a second-order constraint (preventing the ghost instabilities inherent in higher-derivative quantum gravity theories), and a divergence-free identity (satisfying the Bianchi constraint $\nabla_\mu G^{\mu\nu} = 0$ to permit coupling to a conserved source).¹ According to Lovelock's theorem, in exactly four spacetime dimensions, there is only one symmetric, divergence-free, second-order tensor constructible from the metric and its first two derivatives. This definitively forces Einstein-class macroscopic geometry¹:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{(\Psi)}$$

The Dual-Null Source Split and Minimal Action

The macroscopic source tensor $T_{\mu\nu}^{(\Psi)}$ is derived via the exact metric variation of the fundamental loop action. It splits exactly into a dual-null configuration, separating dynamic, finite-excitation configurations from the Lorentz-invariant background¹:

$$T_{\mu\nu}^{(\Psi)} = T_{\mu\nu}^{(NG)} + T_{\mu\nu}^{(bulk)}$$

The corresponding minimal dual action $S_\Psi = S_{NG} + S_{bulk}$ describes the internal structure of an individual closure loop.¹ The Nambu-Goto term governs the kinetic and thermal behaviors of the propagating matter-radiation sector:

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$$S_{NG} = -\sigma_T \int d^2\sigma \sqrt{-h}$$

where σ_T is the worldsheet tension and h is the induced worldsheet metric determinant. Simultaneously, the bulk term describes the unresolvable internal volume of the computational loop, yielding a stress-energy tensor proportional to $g_{\mu\nu}$ and natively deriving the cosmological constant ¹:

$$S_{bulk} = \Lambda_0 \int d^4x \sqrt{-g} \theta_{loop}(x)$$

where Λ_0 is the bulk energy density scale and $\theta_{loop}(x)$ is the loop support function.¹ At macroscopic scales where the system radius R_s far exceeds the string length ℓ_s , higher-order rigidity terms are safely suppressed ($\frac{\hbar}{\sigma_T R_s^2} = (\frac{\ell_s}{R_s})^2 \ll 1$), leaving this minimal dual action as the unique admissible effective theory at low energy.¹

Exact Thermodynamics and the Equation of State

The thermodynamic behavior of the matter sector is governed by the exact Maxwell-Jüttner/Synte interpolation. Defining the dimensionless inverse temperature parameter $z \equiv \frac{m_\Psi c^2}{k_B T_s}$, the equation of state parameter $w(z)$ is derived exactly ¹:

$$w(z) = \frac{1}{z \frac{K_1(z)}{K_2(z)} + 3}$$

where K_1 and K_2 are modified Bessel functions of the second kind. This formulation flawlessly captures the required asymptotic physical behaviors without ad hoc parameterization: for cold, non-relativistic dust ($z \gg 1$), $w \approx 0$; for ultra-relativistic radiation ($z \ll 1$), $w = 1/3$.¹

The vacuum sector, conversely, mandates an equation of state of $w = -1$. The CLG framework establishes this through two independent mathematical routes. Route A relies on symmetry, noting that the Lorentz invariance of the vacuum requires $T_{\mu\nu}^{vac} \propto g_{\mu\nu}$, forcing $p_\Lambda = -\rho_\Lambda$.¹ Route B utilizes a

dynamical bulk proof. The internal energy of the bulk is $U_{bulk} = n_{loop} \Lambda_0 \frac{4\pi}{3} R_0^3 V$. Calculating the bulk pressure via partial derivation with respect to volume yields $p_{bulk} = - \left(\frac{\partial U_{bulk}}{\partial V} \right)_{N, R_0} = -\rho_{bulk}$, identically confirming the $w = -1$ ground-state volumetric bulk tension.¹

Endogenous Micro-Parameter Derivation

The most significant advancement in the updated iteration of the CLG program is the elimination of hand-inserted "Planck-natural" assumptions. Every fundamental micro-parameter is now derived endogenously from the exact intersection between the ontological primitives of Distinguishability (Δ) and Invariance (I).¹

The Loop Mass Scale and Quantum-Gravitational Closure

At the absolute substrate level, a stable closure record Ψ must simultaneously satisfy the Invariance primitive to resist entropic decay while satisfying the Distinguishability primitive to exist as a discrete measurable object.¹ The minimum spatial resolution at which a mass m_Ψ can be localized before its quantum wave-function completely delocalizes is the reduced Compton wavelength:

$$\lambda_C = \frac{\hbar}{m_\Psi c}$$

Conversely, the maximum spatial volume within which this mass can be localized while sustaining structural integrity against total gravitational collapse is defined by the Schwarzschild length:

$$r_S = \frac{Gm_\Psi}{c^2}$$

A stable, minimal computational record demands exact balance. If the mass is lighter, the wave-function delocalizes beyond its own gravitational footprint, violating Invariance. If it is heavier, the geometry undergoes unrecoverable gravitational collapse, violating Distinguishability.¹ The unique mathematical locus where these conditions close without contradiction is $\lambda_C = r_S$:

$$\frac{\hbar}{m_\Psi c} = \frac{Gm_\Psi}{c^2} \implies m_\Psi = \sqrt{\frac{\hbar c}{G}} \equiv m_{Pl}$$

The derived mass m_Ψ is identically the Planck mass. Using the self-consistent 2018 CODATA recommended values for fundamental physical constants—specifically

$$G = 6.67430 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}, c = 299792458 \text{ m/s}, \text{ and}$$

$$\hbar = 1.054571817 \times 10^{-34} \text{ J s}^2 \text{—this calculates exactly to } 2.176434 \times 10^{-8} \text{ kg.}^2$$

The threshold rest energy E_0 , defined as the minimum invariant energy stored within a stationary ground-state loop record, follows immediately via special relativistic equivalence:

$$E_0 = m_\Psi c^2 = \sqrt{\frac{\hbar c^5}{G}} \equiv E_P \approx 1.956 \times 10^9 \text{ J}$$

Internal Degeneracy and String Tension

The degeneracy factor g measures the count of distinct internal quantum states accessible at the threshold energy E_0 . Because the closure record Ψ is the unique stored state of a minimal resolved event, the foundational primitives operate completely below the threshold of composite symmetries. Properties such as fractional spin, electromagnetic charge, and isospin are emergent characteristics of composite loop networks, not properties of the bare ground-state loop. Thus, ontological minimality mandates a scalar ground state, decisively locking the degeneracy at $g = 1$.¹

To derive the fundamental string tension σ_T , the framework relies on the corrected Nambu-Goto mass spectrum. For small transverse deformations $R(\sigma) = R_0 + \xi_n \cos(n\sigma)$ of a bulk-stabilized loop, the squared frequencies are $\omega_n^2 = \frac{n^2}{R_0^2} + m_{bulk}^2$, where the bulk mass gap is defined as $m_{bulk}^2 = \frac{\Lambda_0}{4\sigma_T}$.¹ The CLG program enforces the proper bulk-corrected string mass relation:

$$M^2 c^4 = 4\sigma_T \hbar c \cdot N \left(1 + \frac{\Lambda_0}{4\sigma_T^2 \hbar c \cdot N} \right)$$

As the topological excitation level $N \rightarrow \infty$, the volumetric bulk correction term scales as $1/N$ and vanishes. This crucial insight cleanly repairs earlier linear-energy-step errors present in historical models. The asymptotic ratio $M^2/N \rightarrow 4\sigma_T \hbar c$ rigorously validates the use of Cardy's formula for the two-

dimensional worldsheet conformal field theory ($c_{CFT} = 2$), yielding the exponential Hagedorn density of states $g(E) \sim E^{-a} \exp(E/T_H)$ and a limiting Hagedorn temperature of $T_H = \frac{\hbar c \sqrt{3}}{2\pi R_0}$.¹

For the ground-state closure record, the minimal non-tachyonic topological excitation of the closed worldsheet occurs at $N = 1/4$. At this specific mathematical mode, the volumetric bulk correction identically vanishes. Inserting the derived mass $M = m_{Pl}$ and $N = 1/4$ into the spectrum allows the extraction of the absolute tension:

$$m_{Pl}^2 c^4 = \sigma_T \hbar c \implies \sigma_T = \frac{m_{Pl}^2 c^3}{\hbar} = \frac{c^4}{G} \equiv \sigma_{Pl}$$

Using the CODATA 2018 values, the tension is locked at $\sigma_{Pl} \approx 1.210 \times 10^{44} \text{ N}$.¹ The string tension is not a sliding phenomenological parameter; it intrinsically equals the maximum force permitted by the geometric laws of general relativity.

Vacuum Nucleation, The I-Condition, and the Y-Discriminant

With the micro-parameters definitively endogenized, the framework calculates the physical probability of modern-day loop generation from the vacuum. This nucleation is a Euclidean tunneling process governed by a rate $\Gamma_{create} \sim A_{fluct} \exp(-S_{bounce})$ across the worldsheet manifold.¹

Prior estimates in historical literature erroneously utilized a spherical boundary coefficient of $16\pi^3$, yielding an impossibly large bounce action of $S_{bounce} \sim 10^{256}$. Rigorous alignment with thin-wall Coleman-de Luccia instanton geometry dictates a corrected coefficient factor of $16\pi/3$.¹

$$S_{bounce} = \frac{16\pi}{3} \frac{\sigma_{Pl}^3}{\Lambda_0^2}$$

Given the endogenously derived tension $\sigma_{Pl} \approx 1.210 \times 10^{44} \text{ N}$ and the macroscopic bulk energy density $\Lambda_0 \approx 5.244 \times 10^{-10} \text{ J/m}^3$, the geometrically corrected bounce exponent is calculated exactly to $S_{bounce} \sim 10^{152}$.¹

This exponent establishes the critical Persistence I-Condition. For macroscopic geometry to persist without undergoing rapid spontaneous decay, the loop lifetime $\tau_{loop} \sim \exp(S_{bounce})t_{Pl}$ must comfortably exceed the current age of the universe t_{univ} . This requirement is formalized as:

$$S_{bounce} > \ln \left(\frac{t_{univ}}{t_{Pl}} \right) \approx \ln(8.07 \times 10^{60}) \approx 140$$

With $S_{bounce} \sim 10^{152}$, the I-Condition is satisfied by an overwhelmingly large margin of 152 orders of magnitude against 140.¹ The physical consequence is absolute: present-day loop nucleation from the vacuum is not merely suppressed; it is completely dead. Semiclassical Gaussian prefactors A_{fluct} contribute only logarithmically ($\sim O(1)$) and are mathematically incapable of canceling an exponent of this magnitude.¹ Any theoretical attempt to arbitrarily lower σ_T to Grand Unified Theory (GUT) energy scales would force $S_{bounce} \sim 1$, catastrophically collapsing the loop lifetime to the Planck time and instantly destroying all persistent macroscopic geometry. Thus, dynamical equilibration is falsified, confirming that the universe is permeated by a frozen cosmological relic population forged exclusively in the extreme high-energy epoch of the early universe.¹

The Y-Discriminant and Thermal Falsification

Establishing that the loop gas is a frozen relic condenses the problem of its cosmological origin down to a single mathematically resolvable question: Was this substrate populated via thermal equilibrium dynamics, or through a violent, out-of-equilibrium event? The CLG program resolves this using the parameter-free Y-discriminant.¹

The present-day macroscopic loop number density n_0 is deterministically constrained by cosmological observables and the physical geometry of the loop:

$$n_0 = \frac{\Lambda_{eff}}{\kappa \Lambda_0 V_0}, \quad \text{where} \quad V_0 = \frac{4\pi}{3} R_0^3$$

If this population were produced thermally at a threshold temperature T_{prod} , then after standard Friedmann-Lemaître-Robertson-Walker (FLRW) expansion and dilution, the present density would obey $n_0 = n_{eq}(T_{prod}) \left(\frac{T_0}{T_{prod}} \right)^3$. The non-relativistic equilibrium density is accurately approximated by

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$n_{eq}(T) \approx g \frac{(m_\Psi T)^{3/2}}{(2\pi)^{3/2}} e^{-E_0/T}$. By defining a reduced abundance variable $x \equiv \frac{E_0}{T_{prod}}$ and an aggregated thermal prefactor $A \equiv g T_0^3 \frac{m_\Psi}{(2\pi E_0)^{3/2}}$, the abundance equation compresses elegantly into:

$$n_0 = A x^{3/2} e^{-x}$$

The underlying mathematical function $f(x) = x^{3/2} e^{-x}$ possesses a unique global maximum. Differentiating and setting to zero identifies this maximum exactly at $x = 3/2$. This yields the absolute theoretical upper limit for thermal production, defined as the thermal ceiling constant c^* :

$$c^* \equiv \left(\frac{3}{2}\right)^{3/2} e^{-3/2} \approx 0.40992$$

In a profound phenomenological coincidence, this mathematical constant maps closely to Earth's present axial tilt (obliquity), which oscillates between 22.1° and 24.5° over tens of thousands of years due to planetary perturbations.⁸ At its current value of roughly 23.4366 degrees, the tilt expressed in radians is ≈ 0.40904 rad.¹⁰ The deviation between the Earth's axial tilt in radians and the fundamental thermal ceiling constant c^* is less than 0.2% .¹ While the CLG documents classify this resonance as phenomenological mapping related to broader harmonic geometry frameworks, the critical structural importance of c^* lies in its role in defining the Y-discriminant:

$$Y \equiv \frac{n_0}{A c^*}$$

For any thermal production process to be mathematically permissible within the universe, the inequality $Y \leq 1$ must hold. If $Y < 1$, the exact production temperature T_{prod} can be cleanly recovered utilizing the lower branch of the Lambert W function: $x_{\pm} = -\frac{3}{2} W_{0,-1} \left(-\frac{2}{3} e^{-1} Y^{2/3} \right)$.¹

The exhaustive calculation, utilizing the endogenously derived micro-parameters ($m_\Psi = 2.176 \times 10^{-8} \text{ kg}$, $E_0 = 1.956 \times 10^9 \text{ J}$, $g = 1$, $R_0 = 1.616 \times 10^{-35} \text{ m}$) and the observed macro-parameters ($\Lambda_{eff} = 1.089 \times 10^{-52} \text{ m}^{-2}$, $\Lambda_0 = 5.244 \times 10^{-10} \text{ J/m}^3$, $T_0 = 2.72548 \text{ K}$)¹ produces the following exact outputs:

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Calculation Component	Resulting Value
Present-day loop density (n_0)	$5.654 \times 10^{103} \text{ m}^{-3}$
Thermal Prefactor (A)	$1.256 \times 10^{-94} \text{ m}^{-3}$

Executing the final discriminant logic:

$$Y = \frac{5.654 \times 10^{103}}{(1.256 \times 10^{-94})(0.40992)} \approx 10^{198}$$

Because $Y \approx 10^{198} \gg 1$, the thermal ceiling is exceeded by nearly two hundred orders of magnitude.¹ This represents an absolute mathematical falsification of thermal relic production. To forcefully contort the model to achieve $Y \leq 1$ would require artificially inflating the loop equilibrium radius R_0 to an impossible crossover value $R_{0crit} \approx 1.67 \times 10^{31} \text{ m}$ —a distance exceeding 38,000 times the radius of the observable Hubble horizon.¹ No physically motivated framework can support such an arbitrary geometric distortion. Consequently, the thermal branch (Path 2A) is permanently annihilated, structurally selecting the out-of-equilibrium nonthermal branch (Path 2B) as the singular origin of the universe's loop substrate.¹

Observational Signatures: Gravitational Waves and the Dark Sector

The definitive annihilation of the thermal origin pathway necessitates that the CLG substrate was generated by a specific nonthermal mechanism in the very early universe. The framework identifies three viable out-of-equilibrium sub-branches, each possessing highly distinct, discriminable signatures in the stochastic gravitational-wave background (SGWB)¹:

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Mechanism	Physical Imprint	SGWB Spectral Signature
Kibble / Phase Transition	Condensation from vacuum during symmetry breaking. Network density is topologically set by the correlation length at the transition epoch.	Broken power law: $\Omega_{GW}(f) \propto f^3 [1 + (f/f_{peak})^2]$. Characterized by a causal low-frequency rise with a specific high-frequency tail.
Inflationary Reheating	Inflaton condensate decay violently injects the loop abundance directly into the plasma via parametric resonance.	Broad resonance spectrum with exponential suppression above the peak: $\Omega_{GW}(f) \propto f^3 e^{-f^2/f_{reh}^2}$
Cyclic Pre-Bounce Inheritance	Loop density is inherited strictly as a boundary condition crossing a contracting geometric bottleneck (bouncing cosmology).	Blue-tilted continuum: $\Omega_{GW}(f) \propto f^{n_T}$ where $n_T > 0$. Complemented by squeezed-limit CMB B-mode non-Gaussianity ($f_{NL} \sim 10 - 100$).

Observational discrimination among these pathways relies on exact spectral profiles. The Kibble branch peak frequency is determined by

$$f_{peak}^{(Kibble)} \approx 1.65 \times 10^{-5} \text{ Hz} \cdot (T_K/10^{10} \text{ GeV}) \cdot (g_*(T_K)/106.75)^{1/6}.$$

For the CLG substrate formed at Planck-scale temperatures ($T_K \sim 10^{28}$ GeV), this pushes the primary peak far beyond current detection limits, shifting the optimal observable target to the nanohertz spectral tail accessible by pulsar timing arrays like SKA-PTA.¹ In contrast, the Reheating branch's exponential

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suppression profile is optimally targeted by space-based interferometers like LISA and the terrestrial Einstein Telescope operating in the mHz to kHz bands. Crucially, the discovery of a universally rising blue-tilted tensor continuum ($n_T > 0$) across all observational bands would definitively confirm the Cyclic inheritance mechanism, simultaneously falsifying both the Kibble and Reheating pathways.¹

Beyond the generation of primordial gravitational waves, the CLG substrate presents an inevitable and exact mapping onto the Super-Heavy Dark Matter (SHDM) paradigm.¹ A Planck-mass ($m_{Pl} \sim 10^{19}$ GeV), scalar ($g = 1$), gauge-singlet relic interacts with standard matter exclusively through virtual graviton exchange. The fundamental direct-detection elastic scattering cross-section off a standard nucleon is derived directly from first principles:

$$\sigma_{\Psi N}^{(grav)} \sim \frac{G^2 m_{Pl}^2 m_N^2}{\pi \hbar^4 c^2} \approx 10^{-70} \text{ m}^2$$

This fundamental geometric limit sits 34 orders of magnitude below current, highly sensitive XENON-nT exclusion limits ($\sigma < 10^{-47} \text{ m}^2$). The substrate is maximally dark by mathematical mandate.¹ At the scale of galactic clustering, the vast mass reduces the de Broglie wavelength to $\lambda_{dB} \sim 10^{-32} \text{ m}$ for typical virial velocities, ensuring the clustered fraction of the loop population behaves identically to pressureless cold dark matter on observable macroscopic scales.¹

Information-Geometric Duals: The SHA-256 Isomorphism

The central ontological assertion that physical reality operates as a recursively self-compiling computational substrate strictly implies that mathematically optimized, human-engineered recursive closures should spontaneously exhibit identical structural grammar. The CLG framework identifies the cryptographic hash function SHA-256 not merely as an analogy, but as an exact, deep information-geometric isomorphism to the fundamental physical fold operation.¹

SHA-256 processes data utilizing a 64-round update over eight 32-bit registers (a through h). The specific round update equations perfectly mirror the exact dual-null source split of the CLG macroscopic geometry¹:

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t$$

$$T_2 = \Sigma_0(a) + Maj(a, b, c)$$

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$$a' = T_1 + T_2, \quad e' = d + T_1$$

In this architecture, the T_1 data path utilizes the $Ch(e, f, g)$ bitwise selector. This mathematically represents a nonlinear conditional activation—data flow is either propagated or suppressed based on the immediate local geometry. This maps identically to the S_{NG} Nambu-Goto propagating matter-radiation sector, dictating finite, information-bearing state transitions.¹

Simultaneously, the T_2 data path utilizes the $Maj(a, b, c)$ selector function, a deterministic consensus averaging operator that relentlessly drives the register state toward the background majority. This operates precisely as the S_{bulk} vacuum term—a Lorentz-invariant floor that enforces background stability and completely absorbs localized fluctuations. The final register superposition ($a' = T_1 + T_2$) is the mathematical execution of the dual-null closure, bounded by the compactification of 32-bit modular addition (modulo 2^{32}).¹

This underlying sequence directly maps onto the fundamental five-stage PRESQ cycle inherent to both the CLG fold and the SHA-256 round structure ¹:

PRESQ Stage	CLG Operation	SHA-256 Analogue
P (Position)	Metric configuration $g_{\mu\nu}$	Current register state $(a \dots h)_t$
R (Reflection)	Boundary variation $\delta S / \delta g = 0$	Rotation mixing $\Sigma_0(a), \Sigma_1(e)$
E (Expansion)	Friedmann integration $a(t)$	Message schedule W_t expansion

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S (Synergy)	Sector coupling $Q(\rho_m, \rho_\Lambda)$	$T_1 + T_2$ register merge
Q (Quality)	Y-discriminant audit	Output hash acceptance test

The Lambert W Fixed Point and Carry Propagation

If the isomorphism holds to its deepest levels, the statistical convergence of SHA-256 must match the convergence of the physical substrate. In the CLG framework, a closure event faces a dual-null split with equal a priori weighting (a structural factor of $1/2$). The probability of actualization at the next recursive level $n + 1$ is suppressed exponentially by the audit cost of the previously accumulated state p_n . This generates the fundamental recursive map ¹:

$$p_{n+1} = \frac{1}{2}e^{-p_n}$$

The stable fixed point of this recurrence is uniquely determined by setting $H = (1/2)e^{-H}$, which rearranges algebraically to $He^H = 1/2$. The exact mathematical solution invokes the principal branch of the Lambert W function ¹⁵, denoted as $W_0(z)$ ¹⁷:

$$H = W_0\left(\frac{1}{2}\right) \approx 0.3517337112$$

In standard cryptographic analysis, attention is often paid to the Hamming weight (the fraction of '1' bits), which approaches **0.5** across multiple rounds of SHA-256.¹ However, the Hamming weight merely measures static register content. The CLG framework dictates that the true metric of recursive structural collapse is the *actualization fraction*. In digital modular addition, this corresponds exclusively to the **carry propagation fraction**—the exact percentage of bit-addition operations that successfully generate a cascading carry output rather than a simple un-carried sum.¹ Carries represent actualized, structurally committed closure events bridging bit boundaries.¹⁹

Extensive empirical measurement across the 64 rounds utilizing standard NIST test vectors confirms that the carry generation fraction in SHA-256 strictly converges to $c_{carry} \rightarrow 0.352 \pm 0.008$. This precisely reproduces the Lambert W fixed point $H = W_0(1/2)$ from the first principles of binary modular logic.¹

The factor of $1/2$ stems directly from the underlying base-2 arithmetic, while the exponential decay accurately models the geometric extinction probability of an advancing carry-chain across a 32-bit word boundary.

The profound consequence of this convergence rests in the absolute transcendental nature of the Lambert W function. By the Lindemann-Weierstrass theorem, if H were an algebraic number, e^H would necessarily be transcendental.¹ Yet the relation $He^H = 1/2$ is distinctly rational. If H is algebraic and non-zero, $e^H = 1/(2H)$ must also be algebraic, forcing an unresolvable mathematical contradiction. Therefore, $H = W_0(1/2)$ is strictly transcendental.¹

Cryptographic hardness in SHA-256 is thus proven to not be an arbitrary computational obstacle or an unproven complexity assumption; it is a fundamental consequence of transcendental geometry. Because no algebraic shortcut to a transcendental fixed point exists, no purely algebraic attack can invert the hash fold.¹

Complementing this, the Bailey–Borwein–Plouffe (BBP) formula for π operates as the explicit information-geometric right inverse. Where SHA-256 projects an infinite pre-image space down onto a discrete 256-bit collapsed trace, BBP acts as a directed pointer dereference, unfolding a specific arbitrary position index directly into a resolved transcendental hexadecimal digit without computing preceding states. Restricted to the π -fold manifold, $BBP(n)$ operates as the exact functional inverse $SHA^{-1}(n)$.¹

Cosmological Tension Resolution: The Nexus-Friedmann Control Law

The macroscopic implications of the Lambert W fixed point $H = W_0(1/2)$ extend directly into predictive cosmology, providing a parameter-free resolution to the contemporary H_0 (Hubble parameter) and S_8 (structure growth) cosmological tensions. The framework embeds this recursive logic utilizing the Samson V2 Controller, a Proportional-Integral-Derivative (PID) regulatory architecture functioning mathematically within the Teleparallel Equivalent of General Relativity (TEGR).¹

Unlike arbitrary phenomenological models constructed to fit observational data, the Nexus PID gains are derived entirely and exclusively from the deep geometry of the H -band fixed point¹:

PID Term	Physical Manifestation	Derived Gain Value
Proportional (K_p)	Immediate spacetime elasticity reacting to local metric perturbations.	$K_p = 1$
Integral (K_i)	Accumulated vacuum deviation driving late-time accelerated expansion (Dark Energy).	$K_i = \frac{H}{1-H} \approx 0.5426$
Derivative (K_d)	Anticipatory torsion coupling ($T = -6H_{ubble}^2$) suppressing oscillatory runaway.	$K_d = H \approx 0.3517$

This cybernetic control system operates by actively coupling the universe's matter density (ρ_m) to its vacuum energy density (ρ_Λ). To maintain the critical capacity allocation ratio dictated by H (approximately **35.17%** actualized propagating state versus **64.83%** uncollapsed background potential), the system activates a non-conservative energy transfer term Q exclusively during the late universe, when macroscopic expansion threatens to dilute the actualized matter density below the systemic minimal threshold.¹

This symmetry breaking triggers abruptly at a precise critical redshift z_c , defined strictly by the condition $\Omega_m(z_c) = H$:

$$z_c = \left[\frac{H \cdot \Omega_\Lambda}{\Omega_{m,0}(1-H)} \right]^{1/3} - 1 \approx 0.057$$

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This extremely low redshift ($z < 0.057$, corresponding to roughly the last 800 million years of cosmic time) aligns perfectly with the exact observational window where discrepancies in the Hubble constant between local distance ladders (such as SHoES) and early-universe CMB extrapolations (such as Planck) become irreconcilable under standard Λ CDM assumptions.¹

Within this Nexus-active epoch, the transfer term $Q = K_i H_{ubble} \rho_m$ systematically bleeds energy from the clustered matter sector directly into the dark energy sector. The coupled ODE system integrates the Friedmann expansion alongside this mass-transfer:

$$\dot{\rho}_m + 3H_{ubble} \rho_m = -Q$$

$$\dot{\rho}_\Lambda = +Q$$

Simultaneously, the linear growth factor $D(a)$ for structural fluctuations is modified by the active dilution of the matter background:

$$D'' + (2 - q)H_{ubble} D' - \frac{3}{2}\Omega_m(a)H_{ubble}^2 D = 0$$

The cosmological consequences of this numerical integration are mathematically precise and resolve the primary observational crises in modern astrophysics¹:

1. **Resolution of the H_0 Tension:** The active decay of matter into vacuum energy at late times ($z < 0.057$) effectively increases the local dark energy fraction. This directly enhances the local expansion rate H_0 relative to the high-redshift CMB predictions, systematically driving H_0 upward by an estimated $1.5\% - 3\%$ toward the ~ 73 km/s/Mpc values observed in local supernova surveys.¹
2. **Resolution of the S_8 Tension:** Simultaneously, the accelerated dilution of the matter sector introduces a suppression drag on the linear growth factor of large-scale structure ($D(a)$). This organically damps the clustering amplitude by $10\% - 15\%$, driving the S_8 parameter downward to exactly match the values preferred by late-time weak lensing surveys (e.g., KiDS-1000 and DES Y3).¹

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Crucially, the entire Nexus-Friedmann ODE system possesses exactly zero free parameters beyond standard Λ_{CDM} initial conditions. Every gain and trigger point (K_p, K_i, K_d, z_c) is rigidly locked by the transcendental constant $H = W_0(1/2)$. Furthermore, the Saros cycle of 18.03 years (encompassing 223 synodic months and 242 draconic months)²⁴ echoes the harmonic geometries proposed in the broader Nexus structural mapping, reflecting how deeply the H -band constants permeate scale-invariant causal architectures.¹

Conclusion

The Closure Loop Gas program represents a complete, mathematically constrained realization of a computation-first ontology. By explicitly discarding hand-tuned phenomenological inputs in favor of endogenous derivations drawn from the absolute intersection of quantum distinguishability (λ_c) and gravitational invariance (r_s), the framework successfully isolates the Planck mass, threshold energy, and string tension as unique, irrefutable requirements for structural persistence.

The subsequent recalculation of the Euclidean bounce action utilizing the exact thin-wall Coleman-de Luccia instanton geometry ($S_{\text{bounce}} \sim 10^{152}$) proves that present-day vacuum nucleation is dead, permanently framing the universe as an ancient, frozen cosmological relic. Calculating the Y-discriminant from these newly derived parameters yields an abundance ratio of $Y \approx 10^{198} \gg 1$. This astronomical excess generates an absolute, parameter-free falsification of the thermal origin pathway, restricting cosmology to a highly specific subset of out-of-equilibrium events (such as Cyclic inheritance or Reheating) verifiable via nanohertz to kilohertz stochastic gravitational wave arrays.

Furthermore, the discovery that human-engineered cryptographic hash architectures like SHA-256 autonomously recreate the CLG's exact dual-null macroscopic geometry—and converge precisely upon the same transcendental Lambert W fixed point ($H \approx 0.3517$)—demonstrates a profound isomorphism between applied computer science and theoretical physics. Extrapolated to cosmology via the Nexus-Friedmann equations, this exact fixed point generates a zero-free-parameter PID control architecture capable of simultaneously dissolving the H_0 and S_8 tensions in the modern universe. The framework ultimately shifts the pursuit of quantum gravity away from the search for a new fundamental physical substance, directing it toward the rigorous mathematical analysis of a self-compiling informational geometry constrained solely by the strict limits of recursive admissibility.

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